Yagyaansh Khaneja

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Education

IIT Roorkee, B. Tech. in Eng. Physics (CGPA 7.11/10) *First Division* 2021

RDPS, Delhi (CBSE), Class XII: PCM (94.2 %) 2017

RDPS, Delhi (CBSE), Class X (CGPA 10/10) 2015

Skills

- **Programming:** Python (PyTorch, NumPy, scikit-learn, pandas, matplotlib), FORTRAN, HTML/CSS, SQLite
- **Tools:** MATLAB, COMSOL, CST MW Studio, Excel/Word/PowerPoint, LaTeX, Lightroom
- **Languages:** English (Native), Hindi (Native), French (Basic), Punjabi (Basic)

Achievements

- **NGPE 2019 (Part A):** Top 10%
- **JEE 2017 (Advanced):** AIR 3188
- **JEE 2017 (Mains):** AIR 2933 (Top **0.2**%)
- **Vijyoshi 2016 (Camp):** Invited Guest
- **KYPY 2015 (SA):** AIR 300 (Top **0.4**%)
- **NTSE 2015 (II):** Scholar (Top **0.1**%)

Physics Projects

➢ **Research Intern (Energy Team) – Sonam Wangchuk** [Jan 2022 – May 2022]

- Organization: Himalayan Institute of Alternatives (HIAL), Ladakh
- TL;DR: Assisted renowned environmentalist Sonam Wangchuk on his projects (SDGs)
  - Drafted manual for construction of Passive Solar Heated (PSH) Buildings in Ladakh.
  - Energy Analysis of PCM in Trombé Walls (PSH), and HVAC systems
  - Maintained and installed solar PV systems (20 kW, 7.5 kW)
  - Installed new Stevenson Screen; assisted in weather **data collection** (& analysis) in remote regions of Ladakh using SensorPush and Campbell Scientific weather stations.
- Skills: Research (Thermal Physics), Excel, Project Implementation, Electrical Science

➢ **Bachelor's Project – Silicon Photonic Switch, IIT Roorkee** [May 2020 – Sep 2021]

- Organization: Physics Dept., IIT Roorkee (Prof. Rajesh Kumar)
- TL;DR: Simulated non-volatile GST-SOI waveguides using CST MW Studio and COMSOL.
  - Designed photonic switches for use at 2.1μm (report) and 1 THz (thesis)
  - Studied Mux-DeMux using Optical Phased Array & WDM for telecom applications
- Skills: EM Simulation (COMSOL, CST MW Studio), MATLAB, Waveguide Photonics

➢ **Research Camp – Optics/Photonics, ITMO University** [May 2019 – Jun 2019]

- Organization: ITMO University, St. Petersburg, Russia (Prof. Anton Tcypkin)
- TL;DR: Gained exposure to advances in Optics through active engagement with frontier research (Certificate)
  - Generated THz pulse radiation via laser filamentation of liquids (First global attempt using pH variation of water!)
  - Simulated interference & diffraction patterns for varying waveforms & apertures
  - Studied Quantum Information, Holography, QKD Systems, Femtosecond Lasers, etc.
- Skills: Experimental Optics, MATLAB, Laser Systems

Certificates

- **Machine Learning:** Andrew Ng (Coursera) [2023]
- **Python for Everybody:** Charles Severance (Coursera Specialization) [2020]
- **Web Design for Everybody:** Colleen ven Lent (Coursera Specialization) [2020]
- **Internet: History, Technology, Security:** Charles Severance (Coursera) [2020]

Independent Study (ML)

- **CS50 (Stanford):** Intro to CS [2020]
- **CS50 (Stanford):** AI with Python [2020, 2021]
- **Andrej Karpathy:** NN: Zero to Hero [2024-25]
- **3Blue1Brown:** DL & Transformers [2025]
- **6.S191 (MIT):** Intro to Deep Learning [2025]
- **Kaggle:** Kaggle Learn [2025]

Tech Projects

➢ **Vibe Coding Engineer (Freelance)** [Oct 2025 – present]

- Organization: US based stealth StartUp (San Fransisco)
- TL;DR: Built automation tools for the Hotel industry in the US
  - Wrote playwright scripts (Python) to scrape data from Hotel Management Systems (HMS) like ChoiceAdvantage, IHG Holiday Inn, etc.
  - Built a robust reconciliation engine to find discrepancies in data from different sources (HMS, CLC, Expedia, etc.)
- Results: Got first 2 customers signed up within 1 week of prototype devt.
- Skills: AWS deployment, Processing webpages, data (json, csv) in Python

➢ **Personal Website** – yagyaansh.com [Nov 2020 – Jan 2021]

- TL;DR: Hard-coded my website using basic HTML and CSS (no frameworks)
  - Implemented Dark Theme toggle using JavaScript
  - Wrote blogs with graphics, integrated comments section using Disqus
  - Hosted my website using GitHub Pages
- Skills: Web Dev (HTML, CSS, JS), handling SVG in HTML

Side Projects

➢ **Standard Model - Visualized** [May 2025 – ongoing]

- TL;DR: Visualized Standard Model of Particle Physics in 3D
  - Used three properties at a time (mass, charge, isospin/strangeness)
  - Tried matplotlib initially, then switched to Manim for better animations
- Skills: Python - matplotlib, Manim (by 3Blue1Brown)

➢ **Fee Data Extraction** [Aug 2020]

- TL;DR: API calls (Azure server) to extract data of 3000+ students (college fee receipts)
  - Libraries used - urllib.request, ssl, json, sqlite3 in Python
  - converted the received json data into a SQLite database for easy accessibility
- Skills: Handling JSON and SQLite database using Python

Multidisciplinary Projects

- **Family Business – B2B Wholesale (pan India)** [Sep 2022 – Sep 2025]
- Location: Asia's largest wholesale market: Chandni Chowk/Chawri Bazaar (Delhi-6)
- TL;DR: In essence, increased the efficiency of operations, **saving 4 hours per day**
  - 2022: Migrated **accounts** from legacy (handwritten) to cloud software, enabling remote work for the first time in 35 years
  - 2023: Meticulously classified 1000s of SKUs (**Steel products**), 400+ parties; making data entry organized
  - 2025: Reverse engineered API for a third-party app to automate data entry and reconciliation!
  - 2023-24: Cut operational overhead via warehouse optimization (Streamlined packaging, labels, item codes, colour coded dispatch, and process flows)
  - 2024: Expanded customer base through **online retail** sales via Meesho
  - 2024: Designed a catalog of Door Fittings, Fasteners, Screws - simplifying products for customers (Canva)
  - Learned invaluable lessons about Money, **Cash Flow**, Relationship Management, Business Ethics, Sales, etc.
- Skills: Warehouse Management, Leadership, Product Design, Accounting & Financial Management

- **Leadership Roles, IIT Roorkee** [2017-2021]
  - Additional Secretary, Photography Section
  - Coordinator, Thomso (Cultural Fest)
  - Content Manager, Srishti (Tech Exhibition)
- Skills: Leadership & Management

Community Service

- **Project Manager – Tech Team** [Dec 2024 – Jun 2025]
- Organizations: Tarayana, GIAOS
- TL;DR: Guided the development of data dashboard (PoC)
- Location: Wangdi, Bhutan
  - Empowered the remote village of Rukha in leveraging technology to advance the national vision of Gross National Happiness (GNH)
  - Under the aegis of renowned banker Tan Sri Andrew Sheng SBS
- Skills: Team Management, High Level Architecture Design, (Tech Stack: React Native, MongoDB, DeepSeek API)

- **STEM & English Teacher**
- Location: Ladakh (SECMOL) [2022]
  - Taught dropouts using hands-on science/maths experiments.
  - Led **astronomy** and photography hobby clubs.
- Location: Vietnam (Workaway) [2022]
  - Taught English, Maths to students in the sub-urbs of Hanoi
  - Took part in activities with Sustainable Devt. Community in DaNang
- Location: NSS, IIT Roorkee [2017-2018]
  - Taught English, Science & Maths to children from underprivileged communities around Roorkee
- Skills: Communication, Empathy

Coursework (B. Tech.)

- **Basics**
  - **EMT:** Sadiku; Griffiths
  - **Relativity:** Resnick; Goldstein; Aurther Beiser
  - **QM 1**: Griffiths; Zettili
- **Maths**
  - **Mathematical Physics**: Mary L. Boas; Arfken
  - **Advanced Maths**: Kreyszig; Brown & Churchill
  - **Computational Physics**: DeVries (FORTRAN)
- **Nuclear**
  - **Nuclear Physics**: Krane; John Lilley
  - **Nuclear Astrophysics**: Rolfs & Rodney; Iliadis
- **Interdisciplinary**
  - **Thermal & Statistical Physics:** Sears & Salinger; Zemansky & Dittman; Landau & Lifshitz
  - **Atomic & Molecular Physics**: Haken & Wolf; Bransden
  - **Condensed Matter Physics**: Wahab
  - **Plasma Physics**: Francis F. Chen
- **Electronics**
  - **Electronics**: Malvino; Millman; Streetman; Horowitz; Scherz
  - **Electrical**: Chapman; Hughes
  - **Electrical & Electronic Materials**: S. O. Kasap
  - **Microprocessors**: Ramesh S. Gaonkar
  - **Semiconductor Devices**: Sze & Lee; Streetman
- **Optics/Photonics**
  - **Applied Optics**: Hecht; Born & Wolf; Ghatak
  - **Lasers & Photonics**: Ghatak & Thyagarajan; Silfvast; Svelto; Yariv
  - **Optical Communication**: G. P. Agrawal; Wartak; B. P. Lathi; Schaum
  - **Fiber & Non-Linear Optics**: Ghatak & Thyagarajan
  - **Optoelectronics**: Saleh & Teich; Kasap
  - **Photonic Sensors** (Plasmonics) : Sachin Srivastava; Maeir